

## **UJIAN PRAKTEK BASIS DATA 2**



**Dosen Pengampu:**

**Muhammad Ainurohman, S.Kom**

**Disusun Oleh:**

**Amelia Yanuar 2502004**

**PROGRAM STUDI TEKNIK INFORMATIKA POLITEKNIK  
PURBAYA TEGAL  
2025/2026**

## 1. Analisis Kebutuhan Data Sistem Monitoring Jaringan

### A. Tujuan Sistem

Sistem monitoring jaringan digunakan untuk:

- Memantau perangkat jaringan (router, switch, access point, server)
- Mengetahui status perangkat (online/offline)
- Mencatat penggunaan bandwidth
- Mencatat downtime perangkat
- Mencatat aktivitas teknisi jaringan

### B. Entitas yang dibutuhkan

#### 1) Perangkat (Menyimpan data perangkat jaringan)

Atribut:

- id\_perangkat
- nama\_perangkat
- jenis\_perangkat
- ip\_address
- lokasi
- status

#### 2) Monitoring (Menyimpan hasil monitoring perangkat)

Atribut:

- id\_monitoring
- id\_perangkat
- waktu\_monitoring
- bandwidth\_in
- bandwidth\_out
- uptime
- downtime
- status\_perangkat

#### 3) Teknisi (Menyimpan data teknisi jaringan)

Atribut:

- id\_teknisi
- nama\_teknisi
- no\_hp
- email

#### 4) Aktivitas Teknisi (Mencatat aktivitas teknisi terhadap perangkat)

Atribut:

- id\_aktivitas
- id\_teknisi
- id\_perangkat
- tanggal
- aktivitas
- keterangan

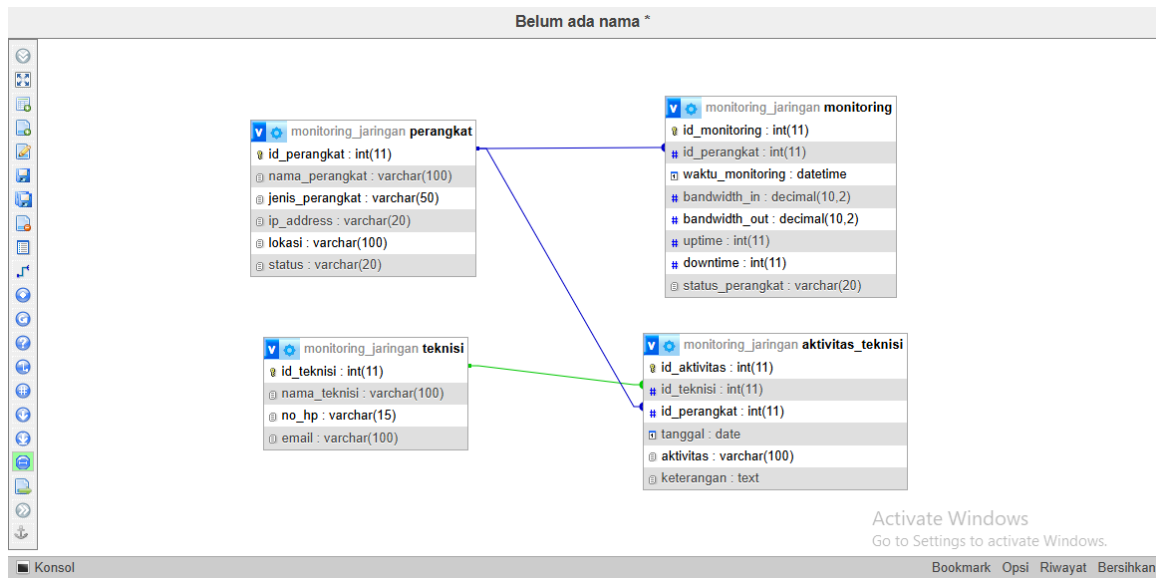
## 2. Entity Relationship Diagram (ERD)

Relasi antar entitas:

- Satu perangkat memiliki banyak monitoring (Perangkat 1 : N Monitoring)
- Satu teknisi dapat melakukan banyak aktivitas (Teknisi 1 : N Aktivitas)

- Satu perangkat dapat memiliki banyak aktivitas teknisi  
(Perangkat 1 : N Aktivitas)

Bentuk ERD:



### 3. Membuat Database dan Tabel Relasional

#### a) Membuat Database

```
Administrator: Command Prompt - mysql -u amel -p
C:\WINDOWS\system32>cd c:\xampp\mysql\bin

c:\xampp\mysql\bin>mysql -u amel -p
Enter password: *****
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| apotek_db |
| information_schema |
| mysql |
| penjualan_toko |
| penjualan_tokoku |
| performance_schema |
| phpmyadmin |
| sistem_event_organizer |
| test |
+-----+
9 rows in set (0.067 sec)

MariaDB [(none)]> CREATE DATABASE monitoring_jaringan;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| apotek_db |
| information_schema |
| monitoring_jaringan |
| mysql |
| penjualan_toko |
| penjualan_tokoku |
| performance_schema |
| phpmyadmin |
| sistem_event_organizer |
+-----+
```

#### b) Membuat Tabel Perangkat

```
Administrator: Command Prompt - mysql -u amel -p
MariaDB [(none)]> use monitoring_jaringan;
Database changed
MariaDB [monitoring_jaringan]> CREATE TABLE perangkat (
  -> id_perangkat INT AUTO_INCREMENT PRIMARY KEY,
  -> nama_perangkat VARCHAR(100),
  -> jenis_perangkat VARCHAR(50),
  -> ip_address VARCHAR(20),
  -> lokasi VARCHAR(100),
  -> status VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.021 sec)
```

### c) Membuat Tabel Monitoring

```
MariaDB [monitoring_jaringan]> CREATE TABLE monitoring (
-> id_monitoring INT AUTO_INCREMENT PRIMARY KEY,
-> id_perangkat INT,
-> waktu_monitoring DATETIME,
-> bandwidth_in DECIMAL(10,2),
-> bandwidth_out DECIMAL(10,2),
-> uptime INT,
-> downtime INT,
-> status_perangkat VARCHAR(20),
-> FOREIGN KEY (id_perangkat)
-> REFERENCES perangkat(id_perangkat)
-> );
Query OK, 0 rows affected (0.049 sec)
```

### d) Membuat Tabel Teknisi

```
MariaDB [monitoring_jaringan]> CREATE TABLE teknisi (
-> id_teknisi INT AUTO_INCREMENT PRIMARY KEY,
-> nama_teknisi VARCHAR(100),
-> no_hp VARCHAR(15),
-> email VARCHAR(100)
-> );
Query OK, 0 rows affected (0.025 sec)
```

### e) Membuat Tabel Aktivitas Teknisi

```
MariaDB [monitoring_jaringan]> CREATE TABLE aktivitas_teknisi (
-> id_aktivitas INT AUTO_INCREMENT PRIMARY KEY,
-> id_teknisi INT,
-> id_perangkat INT,
-> tanggal DATE,
-> aktivitas VARCHAR(100),
-> keterangan TEXT,
-> FOREIGN KEY (id_teknisi)
-> REFERENCES teknisi(id_teknisi),
-> FOREIGN KEY (id_perangkat)
-> REFERENCES perangkat(id_perangkat)
-> );
Query OK, 0 rows affected (0.055 sec)
```

## 4. Implementasi Query DDL dan DML

### a. DDL (Data Definition Language)

- Menambah Kolom

```
MariaDB [monitoring_jaringan]> ALTER TABLE perangkat
-> ADD merk VARCHAR(50);
Query OK, 0 rows affected (0.014 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

- Mengubah Nama Kolom

```
MariaDB [monitoring_jaringan]> ALTER TABLE perangkat
-> CHANGE merk merek VARCHAR(50);
Query OK, 0 rows affected (0.029 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [monitoring_jaringan]> desc perangkat;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id_perangkat | int(11) | NO | PRI | NULL | auto_increment |
| nama_perangkat | varchar(100) | YES | | NULL | |
| jenis_perangkat | varchar(50) | YES | | NULL | |
| ip_address | varchar(20) | YES | | NULL | |
| lokasi | varchar(100) | YES | | NULL | |
| status | varchar(20) | YES | | NULL | |
| merek | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

- Menghapus Kolom

```
MariaDB [monitoring_jaringan]> ALTER TABLE perangkat
-> DROP COLUMN merek;
Query OK, 0 rows affected (0.018 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [monitoring_jaringan]> desc perangkat;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id_perangkat | int(11) | NO | PRI | NULL | auto_increment |
| nama_perangkat | varchar(100) | YES | | NULL | |
| jenis_perangkat | varchar(50) | YES | | NULL | |
| ip_address | varchar(20) | YES | | NULL | |
| lokasi | varchar(100) | YES | | NULL | |
| status | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.020 sec)
```

## b. DML (Data Manipulation Language)

### Insert Data:

#### • Data Perangkat

```
Administrator: Command Prompt - mysql -u amel -p
MariaDB [monitoring_jaringan]> INSERT INTO perangkat
  -> (nama_perangkat, jenis_perangkat, ip_address, lokasi, status)
  -> VALUES
  -> ('Router Utama', 'Router', '192.168.1.1', 'Ruang Server', 'Online'),
  -> ('Switch Lt1', 'Switch', '192.168.1.2', 'Lantai 1', 'Online'),
  -> ('AP Lobby', 'Access Point', '192.168.1.3', 'Lobby', 'Offline');
Query OK, 3 rows affected (0.036 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

#### • Data Monitoring

```
MariaDB [monitoring_jaringan]> INSERT INTO monitoring
  -> (id_perangkat, waktu_monitoring, bandwidth_in,
  -> bandwidth_out, uptime, downtime, status_perangkat)
  -> VALUES
  -> (1, NOW(), 100.50, 90.25, 120, 0, 'Online'),
  -> (2, NOW(), 80.10, 70.00, 100, 5, 'Online'),
  -> (3, NOW(), 0, 0, 0, 50, 'Offline');
Query OK, 3 rows affected (0.008 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

#### • Data Teknisi

```
MariaDB [monitoring_jaringan]> INSERT INTO teknisi
  -> (nama_teknisi, no_hp, email)
  -> VALUES
  -> ('Andi Saputra', '08123456789', 'andi@gmail.com'),
  -> ('Budi Santoso', '08234567891', 'budi@gmail.com');
Query OK, 2 rows affected (0.008 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

#### • Data Aktivitas Teknisi

```
MariaDB [monitoring_jaringan]> INSERT INTO aktivitas_teknisi
  -> (id_teknisi, id_perangkat, tanggal, aktivitas, keterangan)
  -> VALUES
  -> (1, 1, '2025-05-20', 'Maintenance', 'Pengecekan router utama'),
  -> (2, 3, '2025-05-20', 'Perbaikan', 'Access point mati');
Query OK, 2 rows affected (0.007 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

### Data – data yang sudah di input

```
MariaDB [monitoring_jaringan]> select*from perangkat; select*from monitoring;

+----+-----+-----+-----+-----+-----+
| id_perangkat | nama_perangkat | jenis_perangkat | ip_address | lokasi | status |
+----+-----+-----+-----+-----+-----+
| 1 | Router Utama | Router | 192.168.1.1 | Ruang Server | Online |
| 2 | Switch Lt1 | Switch | 192.168.1.2 | Lantai 1 | Online |
| 3 | AP Lobby | Access Point | 192.168.1.3 | Lobby | Offline |
+----+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

+----+-----+-----+-----+-----+-----+-----+-----+
| id_monitoring | id_perangkat | waktu_monitoring | bandwidth_in | bandwidth_out | uptime | downtime | status_perangkat |
+----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 2026-05-21 13:42:06 | 100.50 | 90.25 | 120 | 0 | Online |
| 2 | 2 | 2026-05-21 13:42:06 | 80.10 | 70.00 | 100 | 5 | Online |
| 3 | 3 | 2026-05-21 13:42:06 | 0.00 | 0.00 | 0 | 50 | Offline |
+----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [monitoring_jaringan]> select*from teknisi; select*from aktivitas_teknisi;

+----+-----+-----+-----+
| id_teknisi | nama_teknisi | no_hp | email |
+----+-----+-----+-----+
| 1 | Andi Saputra | 08123456789 | andi@gmail.com |
| 2 | Budi Santoso | 08234567891 | budi@gmail.com |
+----+-----+-----+-----+
2 rows in set (0.000 sec)

+----+-----+-----+-----+-----+-----+
| id_aktivitas | id_teknisi | id_perangkat | tanggal | aktivitas | keterangan |
+----+-----+-----+-----+-----+-----+
| 1 | 1 | 1 | 2025-05-20 | Maintenance | Pengecekan router utama |
| 2 | 2 | 3 | 2025-05-20 | Perbaikan | Access point mati |
+----+-----+-----+-----+-----+-----+
2 rows in set (0.000 sec)
```

### Update Data:

```
Administrator: Command Prompt - mysql -u amel -p
MariaDB [monitoring_jaringan]> UPDATE perangkat
  -> SET status = 'Online'
  -> WHERE id_perangkat = 3;
Query OK, 1 row affected (0.008 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [monitoring_jaringan]> select*from perangkat;

+----+-----+-----+-----+-----+-----+
| id_perangkat | nama_perangkat | jenis_perangkat | ip_address | lokasi | status |
+----+-----+-----+-----+-----+-----+
| 1 | Router Utama | Router | 192.168.1.1 | Ruang Server | Online |
| 2 | Switch Lt1 | Switch | 192.168.1.2 | Lantai 1 | Online |
| 3 | AP Lobby | Access Point | 192.168.1.3 | Lobby | Online |
+----+-----+-----+-----+-----+-----+
3 rows in set (0.000 sec)
```

## Delete Data:

```
MariaDB [monitoring_jaringan]> DELETE FROM monitoring
-> WHERE id_monitoring = 3;
Query OK, 1 row affected (0.005 sec)

MariaDB [monitoring_jaringan]> select*from monitoring;
+-----+-----+-----+-----+-----+-----+-----+-----+
| id_monitoring | id_perangkat | waktu_monitoring | bandwidth_in | bandwidth_out | uptime | downtime | status_perangkat |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 2026-05-21 13:42:06 | 100.50 | 90.25 | 120 | 0 | Online |
| 2 | 2 | 2026-05-21 13:42:06 | 80.10 | 70.00 | 100 | 5 | Online |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.000 sec)
```

## 5. Query Laporan Monitoring Jaringan

### A. Menampilkan Semua Perangkat

```
Administrator: Command Prompt - mysql -u amel -p

MariaDB [monitoring_jaringan]> select*from perangkat;
+-----+-----+-----+-----+-----+-----+
| id_perangkat | nama_perangkat | jenis_perangkat | ip_address | lokasi | status |
+-----+-----+-----+-----+-----+-----+
| 1 | Router Utama | Router | 192.168.1.1 | Ruang Server | Online |
| 2 | Switch Lt1 | Switch | 192.168.1.2 | Lantai 1 | Online |
| 3 | AP Lobby | Access Point | 192.168.1.3 | Lobby | Online |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.000 sec)
```

### B. Menampilkan Perangkat Online

```
MariaDB [monitoring_jaringan]> SELECT nama_perangkat, lokasi, status
-> FROM perangkat
-> WHERE status = 'Online';
+-----+-----+-----+
| nama_perangkat | lokasi | status |
+-----+-----+-----+
| Router Utama | Ruang Server | Online |
| Switch Lt1 | Lantai 1 | Online |
| AP Lobby | Lobby | Online |
+-----+-----+-----+
3 rows in set (0.000 sec)
```

### C. Laporan Monitoring Perangkat

```
MariaDB [monitoring_jaringan]> SELECT
-> p.nama_perangkat,
-> p.jenis_perangkat,
-> m.bandwidth_in,
-> m.bandwidth_out,
-> m.uptime,
-> m.downtime,
-> m.status_perangkat
-> FROM perangkat p
-> JOIN monitoring m
-> ON p.id_perangkat = m.id_perangkat;
+-----+-----+-----+-----+-----+-----+-----+
| nama_perangkat | jenis_perangkat | bandwidth_in | bandwidth_out | uptime | downtime | status_perangkat |
+-----+-----+-----+-----+-----+-----+-----+
| Router Utama | Router | 100.50 | 90.25 | 120 | 0 | Online |
| Switch Lt1 | Switch | 80.10 | 70.00 | 100 | 5 | Online |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

Activate Windows  
Go to Settings

### D. Laporan Aktivitas Teknisi

```
MariaDB [monitoring_jaringan]> SELECT
-> t.nama_teknisi,
-> p.nama_perangkat,
-> a.tanggal,
-> a.aktivitas,
-> a.keterangan
-> FROM aktivitas_teknisi a
-> JOIN teknisi t
-> ON a.id_teknisi = t.id_teknisi
-> JOIN perangkat p
-> ON a.id_perangkat = p.id_perangkat;
+-----+-----+-----+-----+-----+
| nama_teknisi | nama_perangkat | tanggal | aktivitas | keterangan |
+-----+-----+-----+-----+-----+
| Andi Sgputrs | Router Utama | 2025-05-20 | Maintenance | Pengecekan router utama |
| Budi Santoso | AP Lobby | 2025-05-20 | Perbaikan | Access point mati |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

### E. Total Downtime Perangkat

```
MariaDB [monitoring_jaringan]> SELECT
-> p.nama_perangkat,
-> SUM(m.downtime) AS total_downtime
-> FROM monitoring m
-> JOIN perangkat p
-> ON m.id_perangkat = p.id_perangkat
-> GROUP BY p.nama_perangkat;
+-----+-----+
| nama_perangkat | total_downtime |
+-----+-----+
| Router Utama | 0 |
| Switch Lt1 | 5 |
+-----+-----+
2 rows in set (0.001 sec)
```